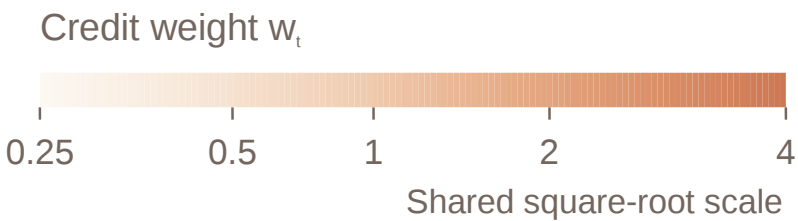


Token-level credit allocation

VPO · Qwen3-14B-Base · $\lambda = 4$



Positive advantage

Deeper orange: stronger reinforcement

(a) Refining an imprecise term

$A = +1.47$ · rank 1/8

Prompt: What are the lasting legacies of the Greek Empire?

The Greek Empire, or more accurately, Ancient Greece, has left a lasting legacy in various aspects of our world today.

(b) Giving concrete guidance

$A = +1.34$ · rank 1/8

Prompt: How can a company promote transparency and accountability?

Encourage leaders to model transparency and accountability, and hold them accountable for their actions and decisions.

Negative advantage

Deeper orange: stronger penalty

(c) Using a misleading category

$A = -1.94$ · rank 8/8

Prompt: What physical factors contribute to the Amazon Rainforest's distinct wildlife?

Tap water: The Amazon Rainforest has a high level of precipitation, ...

(d) Producing an invalid string transformation

$A = -2.16$ · rank 8/8

Prompt: Remove duplicate characters from QFAWjYIbRIJWvLvKLmdNsW.

... here is the output without duplicate characters: QFAWjYIKmsNbvm dw

Reference: QFAWjYIbRJvLKmdNs

CASE (a) / POSITIVE ADVANTAGE

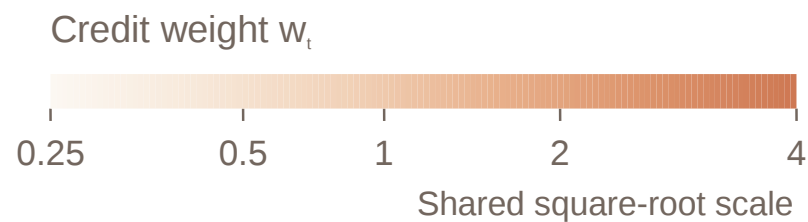
VPO · Qwen3-14B-Base · $\lambda = 4$

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Prompt: What are the lasting legacies of the Greek Empire?

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Ancient Greece, has left a lasting legacy
in various aspects of our world today.



Same saved token weights and shared square-root color scale as the overview.

CASE (b) / POSITIVE ADVANTAGE

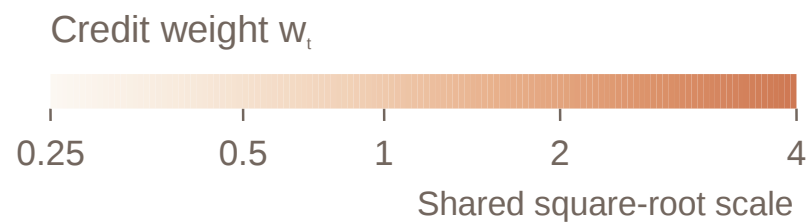
VPO · Qwen3-14B-Base · $\lambda = 4$

(b) Giving concrete guidance

$A = +1.34$ · rank 1/8

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Same saved token weights and shared square-root color scale as the overview.

CASE (c) / NEGATIVE ADVANTAGE

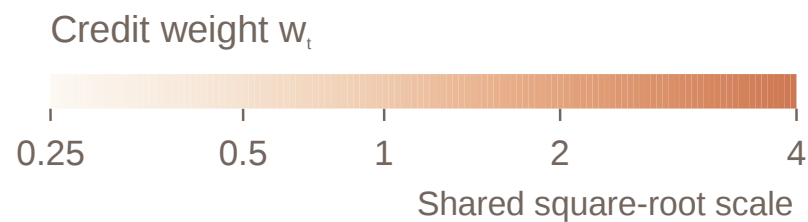
VPO · Qwen3-14B-Base · $\lambda = 4$

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Tap water: The Amazon Rainforest has a high level of precipitation, ...



Same saved token weights and shared square-root color scale as the overview.

CASE (d) / NEGATIVE ADVANTAGE

VPO · Qwen3-14B-Base · $\lambda = 4$

(d) Producing an invalid string transformation

$A = -2.16$ · rank 8/8

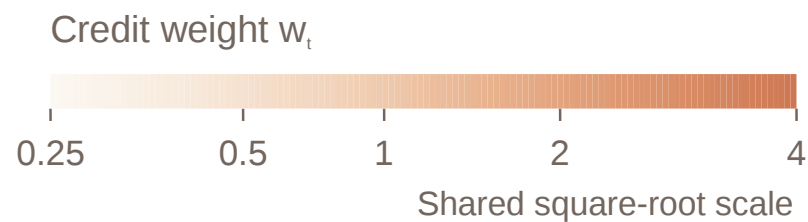
Prompt: Remove duplicate characters
from QFAWjYIbRIJWvLvKLmdNsW.

... here is the output

without duplicate characters:

QFAWjYIKmsNbvm⁴dw

Reference: QFAWjYIbRJvLKmdNs



Same saved token weights and shared square-root color scale as the overview.